

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12411517
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Kim; Heejun et al.

Pedal simulator

Abstract

A pedal simulator is disclosed. The pedal simulator, which is a pedal simulator providing a pedal effort to a pedal of a vehicle, may include: a housing; a piston of which at least a portion of a front is disposed in the housing and which is connected to the pedal to move forward or rearward according to the behavior of the pedal; a displacement member connected to the piston, disposed in the housing, and configured to be displaced together with the piston; and a sensor disposed to detect the displacement of the displacement member.

Inventors:	Kim; Heejun (Gyeonggi-do, KR), Hong; Sungjun (Gyeonggi-do, KR)
Applicant:	HL MANDO CORPORATION (Gyeonggi-do, KR)
Family ID:	93018183
Assignee:	HL MANDO CORPORATION (Gyeonggi-do, KR)
Appl. No.:	18/632233
Filed:	April 10, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240345615 A1	Oct. 17, 2024

Foreign Application Priority Data

KR	10-2023-0047552	Apr. 11, 2023
KR	10-2024-0041106	Mar. 26, 2024

Publication Classification

Int. Cl.: G05G5/03 (20080401); B60T7/04 (20060101); G05G5/05 (20060101)

U.S. Cl.:

CPC **G05G5/03** (20130101); **B60T7/042** (20130101); B60T2220/04 (20130101); G05G5/05 (20130101); G05G2505/00 (20130101)

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2022/0332296	12/2021	Eriksen	N/A	G05G 1/46
2022/0396255	12/2021	Schlangen	N/A	B60T 17/08

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
10-0654335	12/2005	KR	N/A

Primary Examiner: Johnson; Vicky A

Attorney, Agent or Firm: WOMBLE BOND DICKINSON (US) LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority to and the benefit of Korean Patent Application No. 10-2023-0047552, filed on Apr. 11, 2023, Korean Patent Application No. 10-2024-0041106, filed on Mar. 26, 2024, the disclosures of which are incorporated herein by reference in its entirety.

BACKGROUND

- Field of the Invention
- (2) The present disclosure relates to a pedal simulator, and more particularly, to a pedal simulator that provides a pedal effort to a pedal of a vehicle.
- Discussion of Related Art
- (3) Pedal simulators that provide a corresponding reaction force, that is, a pedal effort, when a driver steps on a pedal may be disposed in vehicles. For example, a pedal simulator may be disposed in a brake system of a vehicle to form a brake pedal effort. Generally, a pedal simulator includes components such as a damper, a piston, and a spring to form a pedal effort.
- (4) Conventional pedal simulators only have components for forming a pedal effort and do not have components for detecting a pedal displacement. Accordingly, a separate stroke sensor is mounted on a pedal to detect the movement or displacement of a pedal.
- (5) However, such an external sensor has a structure connected to a pedal of a vehicle through a link and thus has a limitation in that it is actually impossible to secure redundancy. In recent years, redundancy of various sensors has become essential with the development of autonomous vehicle driving technology, and thus development of technology that can overcome structural limitations of the conventional pedal simulators is necessary. (Patent Document) Korean Patent Registration No. 0654335, “Installation structure of pedal stroke sensor,” Date of Registration: Nov. 29, 2006)

SUMMARY OF THE INVENTION

- (6) The present disclosure is directed to providing a pedal simulator having both a component forming a pedal effort and a component detecting a pedal displacement of a vehicle.
- (7) The present disclosure is also directed to providing a pedal simulator that can effectively detect a pedal displacement of a vehicle without being connected to the pedal of the vehicle through a link.
- (8) The present disclosure is also directed to providing a pedal simulator that allows redundancy to be secured in relation to detecting a pedal displacement of a vehicle.
- (9) The present disclosure is also directed to providing a pedal simulator that facilitates assembly in a manufacturing process.
- (10) The objects of the present disclosure are not limited to the above-described objects, and other objects that are not mentioned will be able to be clearly understood by those skilled in the art to which the present disclosure pertains from the following description.
- (11) According to an aspect of the present disclosure, there is provided a pedal simulator providing a pedal effort to a pedal of a vehicle, the pedal simulator including: a housing; a piston of which at least a portion of a front is disposed in the housing and which is connected to the pedal to move forward or rearward according to the behavior of the pedal; a displacement member connected to the piston, disposed in the housing, and displaced together with the piston; and a sensor disposed to detect the displacement of the displacement member.
- (12) In the pedal simulator according to an aspect of the present disclosure, the sensor may be disposed in contact with an outer surface of the housing.
- (13) In the pedal simulator according to an aspect of the present disclosure, the displacement member may include a magnet, and the sensor may detect a change in a magnetic field according to a change in position of the displacement member.
- (14) In the pedal simulator according to an aspect of the present disclosure, the displacement member may further include a holder in which the magnet is disposed and which is coupled to an outer circumferential surface of the piston.
- (15) In the pedal simulator according to an aspect of the present disclosure, the housing may include a first chamber in which the piston is disposed and a second chamber formed parallel to the first chamber and in which the displacement member is disposed.
- (16) In the pedal simulator according to an aspect of the present disclosure, a long groove may be disposed between the first chamber and the second chamber, and the displacement member may be disposed to pass through the long groove.
- (17) In the pedal simulator according to an aspect of the present disclosure, the piston may include a piston head formed in a cylindrical shape and a piston body connected to a rear of the piston head and connected to the pedal.
- (18) In the pedal simulator according to an aspect of the present disclosure, the piston head may include a first outer diameter portion having a first outer diameter, a second outer diameter portion having a second outer diameter larger than the first outer diameter, disposed at a rear of the first outer diameter portion, and forming a support step on a boundary with the first outer diameter portion, and a third outer diameter portion having a third outer diameter larger than the second outer diameter, protruding radially outward, and disposed at a rear of the second outer diameter portion.
- (19) In the pedal simulator according to an aspect of the present disclosure, the piston head may further include a fourth outer diameter portion having a fourth outer diameter smaller than the first outer diameter and disposed at a front of the first outer diameter portion.
- (20) The pedal simulator according to an aspect of the present disclosure may further include a ring-shaped member having an inner diameter corresponding to the first outer diameter, fitted and coupled to the first outer diameter portion, having a rear supported by the support step, and forming a groove, to which the displacement member is able to be fitted and coupled, together with the third

outer diameter portion.

(21) In the pedal simulator according to an aspect of the present disclosure, the ring-shaped member may include a ring-shaped body having an inner diameter corresponding to the first outer diameter and fitted and coupled to the first outer diameter portion, and a flange protruding radially outward from the ring-shaped body.

(22) The pedal simulator according to an aspect of the present disclosure may further include a spring disposed in the housing to be compressed according to forward movement of the piston, and the spring may have a rear seated on the flange to press the flange rearward.

(23) The pedal simulator according to an aspect of the present disclosure may further include a reaction force damper disposed in the housing and configured to provide a reaction force relative to the forward movement of the piston.

(24) According to another aspect of the present disclosure, there is provided a pedal simulator including: a housing; an input shaft connected to a pedal of a vehicle to behave according to the displacement of the pedal; a piston of which at least a portion of a front is disposed in the housing and which is connected to the input shaft to move forward or rearward according to the behavior of the input shaft; a reaction force damper disposed in the housing and configured to provide a reaction force relative to the forward movement of the piston; a displacement member connected to the piston, disposed in the housing, and displaced together with the piston; and a sensor disposed to detect the displacement of the displacement member.

(25) In the pedal simulator according to another aspect of the present disclosure, the sensor may be disposed in contact with an outer surface of the housing.

(26) In the pedal simulator according to another aspect of the present disclosure, the displacement member may include a magnet, and the sensor may detect a change in a magnetic field according to a change in position of the displacement member.

(27) In the pedal simulator according to another aspect of the present disclosure, the displacement member may further include a holder in which the magnet is disposed and which is coupled to an outer circumferential surface of the piston.

(28) In the pedal simulator according to another aspect of the present disclosure, the housing may include a first chamber in which the piston is disposed and a second chamber formed parallel to the first chamber and in which the displacement member is disposed.

(29) In the pedal simulator according to another aspect of the present disclosure, a long groove may be disposed between the first chamber and the second chamber, and the displacement member may be disposed to pass through the long groove.

(30) In the pedal simulator according to another aspect of the present disclosure, the piston may include a piston head formed in a cylindrical shape and a piston body connected to a rear of the piston head and connected to the input shaft, and the piston head may include a first outer diameter portion having a first outer diameter, a second outer diameter portion having a second outer diameter larger than the first outer diameter, disposed at a rear of the first outer diameter portion, and forming a support step on a boundary with the first outer diameter portion, and a third outer diameter portion having a third outer diameter larger than the second outer diameter, protruding radially outward, and disposed at a rear of the second outer diameter portion.

(31) In the pedal simulator according to another aspect of the present disclosure, the piston head may further include a fourth outer diameter portion having a fourth outer diameter smaller than the first outer diameter and disposed at a front of the first outer diameter portion.

(32) The pedal simulator according to another aspect of the present disclosure may further include a ring-shaped member having an inner diameter corresponding to the first outer diameter, fitted and coupled to the first outer diameter portion, having a rear supported by the support step, and forming a groove, to which the displacement member is able to be fitted and coupled, together with the third outer diameter portion.

(33) In the pedal simulator according to another aspect of the present disclosure, the ring-shaped

member may include a ring-shaped body having an inner diameter corresponding to the first outer diameter and fitted and coupled to the first outer diameter portion, and a flange protruding radially outward from the ring-shaped body.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The above and other objects, features and advantages of the present disclosure will become more apparent to those of ordinary skill in the art by describing exemplary embodiments thereof in detail with reference to the accompanying drawings, in which:
- (2) FIG. 1 is a view illustrating a configuration of a brake system of a vehicle in which a pedal simulator according to one embodiment of the present disclosure is disposed;
- (3) FIG. 2 is a perspective view of the pedal simulator according to one embodiment of the present disclosure;
- (4) FIG. 3 is an exploded perspective view of the pedal simulator according to one embodiment of the present disclosure;
- (5) FIG. 4 is a view of a housing of the pedal simulator according to one embodiment of the present disclosure when viewed from the lower side of the front;
- (6) FIG. 5 is a longitudinal cross-sectional view of the pedal simulator according to one embodiment of the present disclosure;
- (7) FIG. 6 is an enlarged view of portion A of FIG. 5; and
- (8) FIG. 7 is a view illustrating a movement of the pedal simulator according to one embodiment of the present disclosure.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS

- (9) Hereinafter, embodiments of the present disclosure will be described in detail so that those skilled in the art to which the present disclosure pertains can easily carry out the embodiments. The present disclosure may be implemented in many different forms and is not limited to the embodiments described herein. In order to clearly describe the present disclosure, portions not related to the description are omitted from the accompanying drawings, and the same or similar components are denoted by the same reference numerals throughout the specification.
- (10) The words and terms used in the specification and the claims are not limitedly construed as their ordinary or dictionary meanings, and should be construed as meaning and concept consistent with the technical spirit of the present disclosure in accordance with the principle that the inventors can define terms and concepts in order to best describe their invention.
- (11) In the specification, it should be understood that the terms such as “comprise” or “have” are intended to specify the presence of features, numbers, steps, operations, components, parts, or combinations thereof described in the specification and do not preclude the possibility of the presence or addition of one or more other features, numbers, steps, operations, components, parts, or combinations thereof.
- (12) FIG. 1 is a view illustrating a configuration of a brake system of a vehicle in which a pedal simulator according to one embodiment of the present disclosure is disposed.
- (13) A pedal simulator **100** is disposed in a vehicle. The pedal simulator **100** provides a corresponding reaction force, that is, a pedal effort, when a driver steps on a pedal **10**.
- (14) Referring to FIG. 1, the pedal simulator **100** may be disposed in a brake system of the vehicle. More specifically, the pedal simulator **100** may be disposed in the brake system of the vehicle and provide a pedal effort to the brake pedal **10**.
- (15) The pedal simulator **100** has both a component forming the pedal effort of the brake pedal **10** and a component detecting a displacement of the brake pedal **10**. Accordingly, the pedal simulator **100** can not only provide a reaction force corresponding to the movement of the brake pedal **10** but

also detect the displacement (movement) of the brake pedal **10**.

(16) Meanwhile, a separate displacement sensor **20** may be disposed near the brake pedal **10**, separately from the pedal simulator **100**, in the brake system of the vehicle. For example, the displacement sensor **20** may be an angle sensor configured to detect an angle of the brake pedal **10**.

(17) According to the present disclosure, since the component detecting the displacement of the brake pedal **10** is disposed in the pedal simulator **100**, and the separate displacement sensor **20** is provided near the brake pedal **10** separately from the pedal simulator **100**, redundancy can be secured in detecting the displacement of the brake pedal **10**.

(18) Meanwhile, displacement information of the brake pedal **10** that is detected by the pedal simulator **100** or the displacement sensor **20** may be transmitted to a pump **30** configured to generate a hydraulic brake pressure. The pump **30** may generate a hydraulic brake pressure that corresponds to the displacement information.

(19) Also, the hydraulic brake pressure generated by the pump **30** may be supplied to a brake **50** disposed near a wheel W of the vehicle through a hydraulic circuit **40**. Braking of the vehicle may be performed as the brake **50** operates due to the hydraulic brake pressure.

(20) Hereinafter, the pedal simulator **100** according to one embodiment of the present disclosure will be described in detail.

(21) FIG. 2 is a perspective view of the pedal simulator according to one embodiment of the present disclosure, and FIG. 3 is an exploded perspective view of the pedal simulator according to one embodiment of the present disclosure. FIG. 4 is a view of a housing of the pedal simulator according to one embodiment of the present disclosure when viewed from the lower side of the front. Also, FIG. 5 is a longitudinal cross-sectional view of the pedal simulator according to one embodiment of the present disclosure.

(22) Referring to FIGS. 2 to 5, the pedal simulator **100** according to one embodiment of the present disclosure may include an input shaft **110**, a housing **120**, a piston **130**, a reaction force damper **140**, a displacement member **150**, a sensor **160**, a ring-shaped member **170**, and a spring **180**.

(23) Hereinafter, in relation to relative positions, the housing **120** will be described as being disposed at a front of the input shaft **110**. That is, a side near the housing **120** will be described as “front,” and a side near the input shaft **110** will be described as “rear.” Also, in relation to movement directions, a direction from the input shaft **110** toward the housing **120** is defined as “forward direction,” and a direction from the housing **120** toward the input shaft **110** is defined as “rearward direction.”

(24) The input shaft **110** is connected to the pedal to behave according to the displacement of the pedal of the vehicle. The input shaft **110** may be disposed to extend to the rear of the housing **120**.

(25) The pedal may be the brake pedal **10** of the vehicle. The input shaft **110** may be connected to the brake pedal **10** to move forward or rearward according to the movement of the brake pedal **10**. More specifically, the input shaft **110** may move forward when the driver of the vehicle steps on the brake pedal **10**.

(26) The housing **120** is provided for the piston **130**, the displacement member **150**, and the like to be disposed therein. In one embodiment of the present disclosure, the housing **120** may include a first chamber **121** and a second chamber **122**.

(27) The first chamber **121** is provided for the reaction force damper **140** and the piston **130** to be disposed therein. The first chamber **121** may have a cylindrical shape. The first chamber **121** may extend in the front-rear direction in the housing **120**.

(28) An inner diameter of the first chamber **121** may be formed to correspond to an outer diameter of the piston **130**. Accordingly, forward movement or rearward movement of the piston **130** may be stably guided in the first chamber **121**.

(29) The second chamber **122** is formed parallel to the first chamber **121** and provided for the displacement member **150** to be disposed therein. The second chamber **122** may have a cylindrical shape. The second chamber **122** may extend in the front-rear direction in the housing **120**. More

specifically, the second chamber **122** may extend in the front-rear direction parallel to the first chamber **121**.

(30) An inner diameter of the second chamber **122** may be formed to have a size that corresponds to the displacement member **150**. Accordingly, forward movement or rearward movement of the displacement member **150** may be stably guided in the second chamber **122**.

(31) Meanwhile, a long groove **123** may be disposed between the first chamber **121** and the second chamber **122**. That is, the first chamber **121** and the second chamber **122** may communicate through the long groove **123**.

(32) The displacement member **150** may be disposed to pass through the long groove **123**. More specifically, the displacement member **150** coupled to the piston **130** may pass through the long groove **123** and be disposed in the second chamber **122**.

(33) Also, the housing **120** may have a sensor arrangement portion **124** for arranging the sensor **160** on an outer surface. The sensor arrangement portion **124** may be disposed on an outer surface of the second chamber **122**. For example, the sensor arrangement portion **124** may have the shape of a plate that provides a seating surface on which the sensor **160** may be seated.

(34) The housing **120** and the sensor arrangement portion **124** may be integrally formed. Of course, in other cases, forming the sensor arrangement portion **124** as a separate member and coupling the sensor arrangement portion **124** to the housing **120** may be considered.

(35) In one embodiment of the present disclosure, the housing **120** has a form having an open front end. Accordingly, components such as the piston **130**, the reaction force damper **140**, and the displacement member **150** may enter the housing **120** through the front of the housing **120** and be disposed therein. In this way, by the housing **120** having the form whose front is open, assembly of the pedal simulator **100** may be easily performed.

(36) In relation to the above, the pedal simulator **100** may further include a housing cover **120a** disposed to cover the open front end of the housing **120**. In one embodiment of the present disclosure, the housing cover **120a** may be coupled to the housing **120** while sealing the open portion at the front end of the housing **120**.

(37) At least a portion of a front of the piston **130** is disposed in the housing **120**. The piston **130** is connected to the input shaft **110** and moves forward or rearward according to the behavior of the input shaft **110**.

(38) Referring to FIG. 3, FIG. 5, and FIG. 6 which is an enlarged view of portion A of FIG. 5, the piston **130** may include a piston head **131** and a piston body **132**.

(39) The piston head **131** is formed in a cylindrical shape. In one embodiment of the present disclosure, the piston head **131** may have a damper groove **131a** formed in an axial direction for the reaction force damper **140** to be inserted therein. The damper groove **131a** may be formed to have a predetermined depth along the longitudinal axis of the piston **130** at a front end of the piston head **131**.

(40) The piston head **131** may include a first outer diameter portion **131c**, a second outer diameter portion **131d**, and a third outer diameter portion **131e**.

(41) The first outer diameter portion **131c** has a cylindrical shape having a first outer diameter. The first outer diameter portion **131c** may have a through-hole that constitutes a portion of the damper groove **131a** along the central axis in the longitudinal direction. The ring-shaped member **170** to be described below may be coupled to an outer circumferential surface of the first outer diameter portion **131c**.

(42) The second outer diameter portion **131d** has a second outer diameter larger than the first outer diameter and is disposed at a rear of the first outer diameter portion **131c**. The second outer diameter portion **131d** may have a through-hole that constitutes a portion of the damper groove **131a** along the central axis in the longitudinal direction.

(43) The second outer diameter portion **131d** forms a support step S2 on a boundary with the first outer diameter portion **131c**. The ring-shaped member **170**, which will be described in detail below,

may be supported by the support step S2.

(44) The third outer diameter portion **131e** has a third outer diameter larger than the second outer diameter, protrudes radially outward, and is disposed at a rear of the second outer diameter portion **131d**. The third outer diameter portion **131e** may have a through-hole that constitutes a portion of the damper groove **131a** along the central axis in the longitudinal direction.

(45) The third outer diameter portion **131e** may form a groove G in an outer circumferential surface of the piston head **131** together with the ring-shaped member **170**. More specifically, the third outer diameter portion **131e** of the piston head **131** may form the groove G in a circumferential direction of the piston head **131** together with the ring-shaped member **170**.

(46) The displacement member **150** may be coupled to the groove G formed by the third outer diameter portion **131e** and the ring-shaped member **170**.

(47) In one embodiment of the present disclosure, the piston head **131** may further include a fourth outer diameter portion **131b**.

(48) The fourth outer diameter portion **131b** has a fourth outer diameter smaller than the first outer diameter and is disposed at a front of the first outer diameter portion **131c**. The ring-shaped member **170** may enter toward the front of the piston head **131** and be coupled to the first outer diameter portion **131c**. The fourth outer diameter portion **131b** facilitates entry of the ring-shaped member **170** toward the first outer diameter portion **131c**.

(49) A sloped portion S1 is formed between the fourth outer diameter portion **131b** and the first outer diameter portion **131c**. Here, the sloped portion S1 may be formed so that an outer diameter thereof gradually increases from the fourth outer diameter portion **131b** toward the first outer diameter portion **131c**. When the sloped portion S1 is formed to have an outer diameter gradually increasing from the front toward the rear in this way, during assembly of the ring-shaped member **170**, entry of the ring-shaped member **170** from the fourth outer diameter portion **131b** toward the first outer diameter portion **131c** may be smoothly guided.

(50) The piston body **132** is connected to the rear of the piston head **131**. The piston body **132** is connected to the input shaft **110**. The piston body **132** may have a cylindrical shape. The piston body **132** may have a predetermined outer diameter and extend to the rear of the piston head **131**.

(51) Referring to FIG. 5, the piston body **132** may have an input shaft groove **132a** formed to have a predetermined depth from a rear end toward the front along the central axis in the longitudinal direction. In one embodiment of the present disclosure, the damper groove **131a** and the input shaft groove **132a** do not communicate with each other.

(52) The input shaft **110** is disposed to be inserted into the input shaft groove **132a**. More specifically, a front portion of the input shaft **110** may be disposed to be inserted into the input shaft groove **132a**. A front sidewall of the input shaft groove **132a** may be disposed in contact with a front end of the input shaft **110**, and when the input shaft **110** moves forward according to the movement of the brake pedal **10**, the input shaft **110** may press the front sidewall of the input shaft groove **132a**, and accordingly, the piston **130** may move forward.

(53) In one embodiment of the present disclosure, the piston body **132** has an extending portion **132c** that has a predetermined outer diameter and extends in a cylindrical shape. The above-described input shaft groove **132a** may be formed along the central axis in the longitudinal direction of the extending portion **132c**.

(54) In one embodiment of the present disclosure, the piston body **132** may further include a diameter reduction portion **132b** that is recessed radially inward at a front portion of the extending portion **132c**. A support ring **190** may be coupled to the diameter reduction portion **132b**. The support ring **190** may prevent a rear portion of the piston head **131** from directly coming in contact with a wall body of the housing **120** and may absorb impact.

(55) The reaction force damper **140** is disposed in the housing **120** and provides a reaction force relative to the forward movement of the piston **130**. The reaction force damper **140** provides a force that resists the forward movement of the piston **130**, and in this way, the driver of the vehicle may

feel a pedal sensation.

(56) The reaction force damper **140** may have any structure that provides an elastic restoring force for resisting the forward movement of the piston **130**. Also, the reaction force damper **140** may be formed of a member made of a single material having elasticity.

(57) In one embodiment of the present disclosure, a front end of the reaction force damper **140** is supported by the housing cover **120a**, and a rear end of the reaction force damper **140** is supported by the piston **130**. More specifically, a rear portion of the reaction force damper **140** may be disposed to be inserted into the damper groove **131a** of the piston head **131**. Here, the rear end of the reaction force damper **140** may be supported by a sidewall of the damper groove **131a**.

(58) The displacement member **150** is connected to the piston **130**, disposed in the housing **120**, and displaced together with the piston **130**. The position of the piston **130** may be identified by detecting a change in the position of the displacement member **150**. Since the displacement of the piston **130** occurs corresponding to the displacement of the input shaft **110** connected to the pedal of the vehicle, the displacement of the pedal may be identified by identifying the position of the piston **130**.

(59) The displacement member **150** may include a magnet **151**. By the displacement member **150** including the magnet **151** in this way, the sensor **160** to be described below may detect a change in a magnetic field according to a change in the position of the displacement member **150**.

(60) The displacement member **150** may further include a holder **152** in which the magnet **151** is disposed and which is coupled to an outer circumferential surface of the piston **130**. The holder **152** allows the magnet **151** to be coupled to the piston **130** for the magnet **151** to move forward or rearward together with the piston **130** according to the forward or rearward movement of the piston **130**.

(61) Referring to FIG. 3, the holder **152** may include a coupling portion **152a** and a magnet arrangement portion **152b**.

(62) The coupling portion **152a** is a portion coupled to the piston **130**. In one embodiment of the present disclosure, the coupling portion **152a** may be fitted and coupled to the outer circumferential surface of the piston head **131** of the piston **130**.

(63) As described above, in a state in which the ring-shaped member **170** is coupled to the piston head **131**, the ring-shaped member **170** and the third outer diameter portion **131e** of the piston head **131** form the groove G in the circumferential direction of the piston head **131**. The coupling portion **152a** may have an arc shape inserted into and coupled to the groove G.

(64) The magnet arrangement portion **152b** is provided for the magnet **151** to be disposed therein. The magnet arrangement portion **152b** is connected to the coupling portion **152a**.

(65) In one embodiment of the present disclosure, the magnet arrangement portion **152b** may be formed of a cylindrical member in which the magnet **151** is disposed. More specifically, the magnet arrangement portion **152b** may have an outer diameter that corresponds to the inner diameter of the second chamber **122**.

(66) The magnet arrangement portion **152b** may extend in the front-rear direction in the second chamber **122**. Here, the magnet arrangement portion **152b** may extend in a direction orthogonal to the arc-shaped coupling portion **152a**.

(67) Meanwhile, the holder **152** may further have one or more protrusions **152c** that protrude toward the ring-shaped member **170** in order to come in contact with the ring-shaped member **170**. The protrusions **152c** form a contact portion between the displacement member **150** and the ring-shaped member **170** and help the displacement member **150** remain stably fixed without a clearance in the groove G. A plurality of protrusions **152c** may be symmetrically provided.

(68) The sensor **160** is disposed to detect the displacement of the displacement member **150**. The sensor **160** may detect a change in the magnetic field according to the forward or rearward movement of the magnet **151** of the displacement member **150**. For example, the sensor **160** may be a Hall sensor.

(69) The input shaft **110** and the piston **130** move forward or rearward according to the displacement of the pedal of the vehicle, and the displacement member **150** moves forward or rearward together with the piston **130** according to the forward or rearward movement of the piston **130**. In this process, a change in the magnetic field occurs according to the displacement of the magnet **151** of the displacement member **150**.

(70) The sensor **160** may detect a change in the magnetic field to detect the position of the displacement member **150**. The displacement (position) of the pedal of the vehicle may be measured based on the information detected by the sensor **160** in this way.

(71) The sensor **160** may be disposed in contact with an outer surface of the housing **120**. In one embodiment of the present disclosure, the sensor **160** may be disposed in contact with an outer surface of the second chamber **122** of the housing **120**. More specifically, the sensor **160** may be coupled to the sensor arrangement portion **124** provided in contact with the outer surface of the second chamber **122** of the housing **120**.

(72) Referring to FIG. 3, the sensor **160** may be fastened to the outer surface of the housing **120**, that is, the outer surface of the sensor arrangement portion **124**, through a bolt B. Also, the sensor **160** may have a connection interface **160a** for supplying power and transmitting detected information.

(73) The ring-shaped member **170** is coupled to the piston **130** and forms the groove G that allows the displacement member **150** to be coupled to the piston **130**. The ring-shaped member **170** may have an inner diameter that corresponds to the first outer diameter and may be fitted and coupled to the first outer diameter portion **131c**.

(74) Referring to FIGS. 5 and 6, the ring-shaped member **170** has an inner diameter that corresponds to the first outer diameter, is fitted and coupled to the first outer diameter portion **131c**, has a rear supported by the support step S2, and forms the groove G, to which the displacement member **150** is able to be fitted and coupled, together with the third outer diameter portion **131e**.

(75) In one embodiment of the present disclosure, the ring-shaped member **170** may include a ring-shaped body **171** and a flange **172**.

(76) The ring-shaped body **171** has an inner diameter that corresponds to the first outer diameter. The ring-shaped body **171** may be fitted and coupled to the first outer diameter portion **131c**. Here, the ring-shaped body **171** may enter toward the front of the piston head **131** and be coupled to the first outer diameter portion **131c**.

(77) As described above, the fourth outer diameter portion **131b** having the fourth outer diameter smaller than the first outer diameter of the first outer diameter portion **131c** is disposed at the front of the first outer diameter portion **131c** of the piston head **131**. Since the fourth outer diameter portion **131b** has an outer diameter smaller than the inner diameter of the ring-shaped body **171** of the ring-shaped member **170**, the fourth outer diameter portion **131b** facilitates entry of the ring-shaped body **171** toward the first outer diameter portion **131c**.

(78) Also, the sloped portion S1 is formed between the fourth outer diameter portion **131b** and the first outer diameter portion **131c**, and the sloped portion S1 may have a shape whose outer diameter gradually increases from the fourth outer diameter portion **131b** toward the first outer diameter portion **131c**. Such a configuration allows the ring-shaped body **171** to be smoothly guided from the fourth outer diameter portion **131b** toward the first outer diameter portion **131c** during assembly of the ring-shaped member **170**.

(79) The flange **172** protrudes radially outward from the ring-shaped body **171**. The flange **172** may form the groove G, to which the displacement member **150** is able to be fitted and coupled, together with the third outer diameter portion **131e**.

(80) In the pedal simulator **100** according to one embodiment of the present disclosure, the groove G to which the displacement member **150** is able to be coupled is formed through a combination of the ring-shaped member **170** and the third outer diameter portion **131e** of the piston head **131**. That is, a structure in which a portion of the piston head **131** is formed to be recessed radially inward is

provided through the combination of the ring-shaped member **170** and the third outer diameter portion **131e**.

(81) Such a configuration allows assembly of the pedal simulator **100** to be performed by placing the piston **130** in the first chamber **121** of the housing **120**, placing the displacement member **150** on the piston head **131**, and then fitting and coupling the ring-shaped member **170** to the piston head **131**. Accordingly, it is easy to place the piston **130** and the displacement member **150** in the housing **120**, and efficiency can be secured in terms of assembly and manufacture.

(82) The spring **180** is disposed in the housing **120** to be compressed according to the forward movement of the piston **130**. The spring **180** may have a cylindrical shape.

(83) The spring **180** contributes to forming a pedal sensation along with the reaction force damper **140**. Also, the spring **180** may serve to move the piston **130** rearward for the pedal to return to its original position when a force stepping on the pedal is released.

(84) In one embodiment of the present disclosure, the spring **180** may have a rear seated on the flange **172** to press the flange **172** rearward. In other words, the spring **180** presses the ring-shaped member **170** toward the support step **S2** formed between the first outer diameter portion **131c** and the second outer diameter portion **131d**. In this way, a coupling state between the ring-shaped member **170** and the piston **130** may be stably maintained.

(85) FIG. 7 is a view illustrating a movement of the pedal simulator according to one embodiment of the present disclosure.

(86) Referring to FIG. 7, when the driver of the vehicle steps on the pedal, the pedal is displaced forward, and the input shaft **110** connected to the pedal moves forward. Accordingly, the piston **130** connected to the input shaft **110** moves forward. Here, the displacement member **150** coupled to the piston **130** moves forward together with the piston **130**.

(87) As described above, the displacement member **150** includes the magnet **151**. Therefore, when the displacement member **150** moves forward, the magnet **151** moves forward together, and as a result, a magnetic field around the displacement member **150** changes.

(88) The sensor **160** may detect such a change in the magnetic field. From the change in the magnetic field detected by the sensor **160**, the position of the displacement member **150** may be determined, and the positions of the piston **130** connected to the displacement member **150** and the pedal of the vehicle may be determined.

(89) Control of the vehicle may be performed according to the displacement (position) of the pedal of the vehicle that is determined as described above. For example, when the pedal is the brake pedal **10** disposed in the brake system of the vehicle, the magnitude of the brake force that corresponds to the displacement (position) of the brake pedal **10** may be determined. The pump **30**, which is configured to generate a hydraulic brake pressure according to the determined magnitude of the brake force, may generate a hydraulic brake pressure that corresponds thereto and may supply the hydraulic brake pressure to the brake **50** disposed at each wheel **W** through the hydraulic circuit **40** of the vehicle.

(90) According to the above-described configurations, the pedal simulator according to an aspect of the present disclosure has both a component forming a pedal effort and a component detecting a pedal displacement of a vehicle and thus can efficiently detect the displacement of the pedal.

(91) The pedal simulator according to an aspect of the present disclosure can effectively detect a pedal displacement of a vehicle without being connected to the pedal of the vehicle through a link.

(92) The pedal simulator according to an aspect of the present disclosure can be used together with a separate external sensor and thus allows redundancy to be secured in relation to detecting a pedal displacement of a vehicle.

(93) The pedal simulator according to an aspect of the present disclosure provides high assembly efficiency in a manufacturing process through an open front structure of a housing, a stepped structure of a piston, and the like.

(94) It should be understood that the effects of the present disclosure are not limited to the above-

described effects, and include all effects inferable from a configuration of the invention described in detailed descriptions or claims of the present disclosure.

(95) Although embodiments of the present disclosure have been described, the spirit of the present disclosure is not limited by the embodiments presented in the specification. Those skilled in the art who understand the spirit of the present disclosure will be able to easily suggest other embodiments by adding, changing, deleting, or adding components within the scope of the same spirit, but this will also be included within the scope of the spirit of the present disclosure.

Claims

1. A pedal simulator providing a pedal effort to a pedal of a vehicle, the pedal simulator comprising: a housing; a piston of which at least a portion of a front is disposed in the housing and which is connected to the pedal to move forward or rearward according to the behavior of the pedal; a displacement member connected to the piston, disposed in the housing, and displaced together with the piston; and a sensor disposed to detect the displacement of the displacement member, wherein: the displacement member includes a magnet; and the sensor detects a change in a magnetic field according to a change in position of the displacement member.
2. The pedal simulator of claim 1, wherein the sensor is disposed in contact with an outer surface of the housing.
3. The pedal simulator of claim 1, wherein the displacement member further includes a holder in which the magnet is disposed and which is coupled to an outer circumferential surface of the piston.
4. The pedal simulator of claim 1, wherein the housing includes: a first chamber in which the piston is disposed; and a second chamber formed parallel to the first chamber and in which the displacement member is disposed.
5. The pedal simulator of claim 4, wherein a through hole is disposed between the first chamber and the second chamber, and the displacement member is disposed to pass through the long groove.
6. The pedal simulator of claim 1, wherein: the piston includes a piston head formed in a cylindrical shape and a piston body connected to a rear of the piston head and connected to the pedal; and the piston head includes a first outer diameter portion having a first outer diameter, a second outer diameter portion having a second outer diameter larger than the first outer diameter, disposed at a rear of the first outer diameter portion, and forming a support step on a boundary with the first outer diameter portion, and a third outer diameter portion having a third outer diameter larger than the second outer diameter, protruding radially outward, and disposed at a rear of the second outer diameter portion.
7. The pedal simulator of claim 6, wherein the piston head further includes a fourth outer diameter portion having a fourth outer diameter smaller than the first outer diameter and disposed at a front of the first outer diameter portion.
8. The pedal simulator of claim 6, further comprising a ring-shaped member having an inner diameter corresponding to the first outer diameter, fitted and coupled to the first outer diameter portion, having a rear supported by the support step, and forming a groove, to which the displacement member is fitted and coupled, together with the third outer diameter portion.
9. The pedal simulator of claim 8, wherein the ring-shaped member includes: a ring-shaped body having an inner diameter corresponding to the first outer diameter and fitted and coupled to the first outer diameter portion; and a flange protruding radially outward from the ring-shaped body.
10. The pedal simulator of claim 9, further comprising a spring disposed in the housing to be compressed according to forward movement of the piston, wherein the spring has a rear seated on the flange to press the flange rearward.
11. A pedal simulator comprising: a housing; an input shaft connected to a pedal of a vehicle to behave according to the displacement of the pedal; a piston of which at least a portion of a front is

disposed in the housing and which is connected to the input shaft to move forward or rearward according to the behavior of the input shaft; a reaction force damper disposed in the housing and configured to provide a reaction force relative to the forward movement of the piston; a displacement member connected to the piston, disposed in the housing, and displaced together with the piston; and a sensor disposed in contact with an outer surface of the housing and disposed to detect the displacement of the displacement member, wherein: the displacement member includes a magnet; and the sensor detects a change in a magnetic field according to a change in position of the displacement member.

12. The pedal simulator of claim 11, wherein the displacement member further includes a holder in which the magnet is disposed and which is coupled to an outer circumferential surface of the piston.

13. The pedal simulator of claim 11, wherein the housing includes: a first chamber in which the piston is disposed; and a second chamber formed parallel to the first chamber and in which the displacement member is disposed.

14. The pedal simulator of claim 13, wherein a through hole is disposed between the first chamber and the second chamber, and the displacement member is disposed to pass through the long groove.

15. The pedal simulator of claim 11, wherein: the piston includes a piston head formed in a cylindrical shape and a piston body connected to a rear of the piston head and connected to the input shaft; and the piston head includes a first outer diameter portion having a first outer diameter, a second outer diameter portion having a second outer diameter larger than the first outer diameter, disposed at a rear of the first outer diameter portion, and forming a support step on a boundary with the first outer diameter portion, and a third outer diameter portion having a third outer diameter larger than the second outer diameter, protruding radially outward, and disposed at a rear of the second outer diameter portion.

16. The pedal simulator of claim 15, wherein the piston head further includes a fourth outer diameter portion having a fourth outer diameter smaller than the first outer diameter and disposed at a front of the first outer diameter portion.

17. The pedal simulator of claim 15, further comprising a ring-shaped member having an inner diameter corresponding to the first outer diameter, fitted and coupled to the first outer diameter portion, having a rear supported by the support step, and forming a groove, to which the displacement member is fitted and coupled, together with the third outer diameter portion.

18. The pedal simulator of claim 17, wherein the ring-shaped member includes: a ring-shaped body having an inner diameter corresponding to the first outer diameter and fitted and coupled to the first outer diameter portion; and a flange protruding radially outward from the ring-shaped body.
